INCH-POUND

AN848 Rev 13
w/AMENDMENT 2
21 June 2023
SUPERSEDING
AN848 Rev 13
w/AMENDMENT 1
20 February 2020

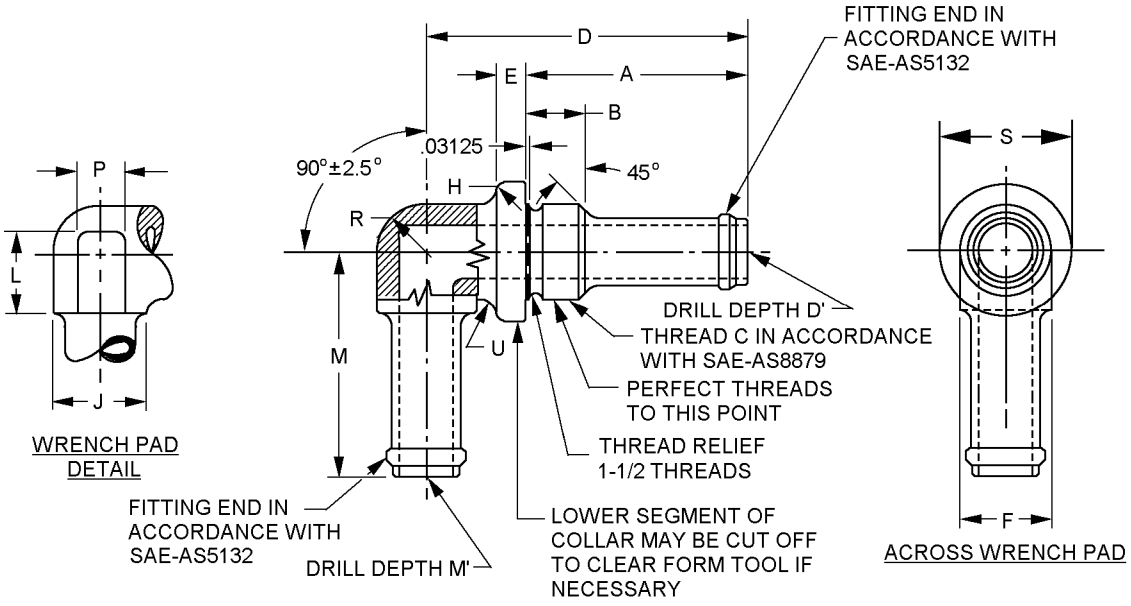
DETAIL SPECIFICATION SHEET

ELBOW, HOSE, 90 DEGREE

Reinstated after 7 June 2011. Inactive for new design.
For new design, use SAE-AS5188.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4843.



Inch	mm
.01325	0.337

See notes at end of figure.

FIGURE 1. 90° elbow dimensions and configuration.

AMSC N/A

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FSC 4730

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Dash number	Hose ID inches (mm)	A inches (mm)	B inches (mm)	Thread C Ref SAE-AS8879	D +.047 (1.19) -.000 inches (mm)
-4	.250 (6.35)	1.938 (49.23)	.438 (11.13)	.4375-20UNJF-3A	2.438 (61.92)
-6	.375 (9.53)	1.969 (50.01)	.469 (11.91)	.5625-18UNJF-3A	2.625 (66.68)
-8	.500 (12.70)	2.000 (50.80)	.500 (12.70)	.7500-16UNJF-3A	2.781 (70.64)
-10	.675 (17.15)	2.031 (51.59)	.531 (13.49)	1.875 -12UNJ-3A	2.969 (75.41)
-12	.750 (19.05)	2.063 (52.40)	.563 (14.30)	1.0625 -12UNJ-3A	3.094 (78.59)
-16	1.000 (25.40)	2.063 (52.40)	.563 (14.30)	1.3125-12UNJ-3A	3.219 (81.76)
-20	1.250 (31.75)	2.063 (52.40)	.563 (14.30)	1.6250-12UNJ-3A	3.531 (89.69)
-24	1.500 (38.10)	2.063 (52.40)	.563 (14.30)	1.8750-12UNJ-3A	3.656 (92.86)

Dash number	D' +.047 (1.19) -.000 inches (mm)	E inches (mm)	F inches (mm)	H Radius inches (mm)	J Diameter inches (mm)	L Approx. inches (mm)
-4	2.500 (63.50)	.156 (3.96)	.438 (11.13)	.063 (1.60)	.438 (11.13)	.313 (7.95)
-6	2.672 (67.87)	.188 (4.78)	.563 (14.30)	.063 (1.60)	.563 (14.30)	.500 (12.70)
-8	2.844 (72.24)	.219 (5.56)	.750 (19.05)	.094 (2.39)	.750 (19.05)	.625 (15.88)
-10	3.094 (78.59)	.250 (6.35)	.875 (22.23)	.094 (2.39)	.875 (22.23)	.750 (19.05)
-12	3.219 (81.76)	.250 (6.35)	1.063 (27.00)	.094 (2.39)	1.063 (27.00)	.938 (23.83)
-16	3.344 (84.94)	.250 (6.35)	1.313 (33.35)	.125 (3.18)	1.313 (33.35)	1.000 (25.40)
-20	3.656 (92.86)	.281 (7.14)	1.625 (41.28)	.125 (3.18)	1.625 (41.28)	1.125 (28.58)
-24	3.781 (96.04)	.281 (7.14)	1.875 (47.63)	.125 (3.18)	1.875 (47.63)	1.500 (38.10)

Dash number	M +.047 (1.19) -.000 inches (mm)	M' +.047 (1.19) -.000 inches(mm)	P Approx inches(mm)	R Radius inches (mm)	S Diameter inches (mm)	U Radius inches (mm)
-4	1.781 (45.24)	1.828 (46.43)	.250 (6.35)	.219 (5.56)	.625 (15.88)	.063 (1.60)
-6	1.875 (47.63)	1.923 (48.84)	.313 (7.95)	.281 (7.14)	.813 (20.65)	.094 (2.39)
-8	1.969 (50.01)	2.031 (51.59)	.438 (11.13)	.375 (9.53)	1.000 (25.40)	.094 (2.39)
-10	2.063 (52.40)	2.188 (55.58)	.438 (11.13)	.438 (11.13)	1.250 (31.75)	.125 (3.18)
-12	2.156 (54.76)	2.281 (57.94)	.500 (12.70)	.531 (13.49)	1.438 (36.53)	.125 (3.18)
-16	2.281 (57.94)	2.406 (61.11)	.563 (14.30)	.656 (16.66)	1.688 (42.88)	.125 (3.18)
-20	2.500 (63.50)	2.625 (66.68)	.563 (14.30)	.813 (20.65)	2.063 (52.40)	.188 (4.48)
-24	2.625 (66.68)	2.750 (69.85)	.625 (15.88)	.938 (23.83)	2.375 (60.33)	.188 (4.48)

Max weight lbs (kg)			
Copper	Al	CRES	Ti
.120 (0.05)	.040 (0.02)	.120 (0.05)	.066 (0.03)
.180 (0.08)	.060 (0.03)	.180 (0.08)	.100 (0.05)
.300 (0.14)	.100 (0.05)	.300 (0.14)	.165 (0.07)
.390 (0.17)	.130 (0.06)	.389 (0.18)	.215 (0.10)
.600 (0.27)	.200 (0.09	.599 (0.27)	.331 (0.15)
.870 (0.39)	.290 (0.13)	.868 (0.39)	.479 (0.22)
1.500 (0.68)	.500 (0.23)	1.497 (0.68)	.827 (0.38)
2.100 (0.95)	.700 (0.32)	2.096 (0.95)	1.157 (0.52)

See notes at end of figure.

FIGURE 1. 90° elbow dimensions and configuration - Continued.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.016 inch (0.41 mm).
4. Break sharp edges .and remove all hanging burrs and slivers
5. Machined surfaces shall be finished to 125µ in Ra, forged surfaces shall be 250µ inches Ra, unless otherwise specified on the figures. Surface finish shall be in accordance with ASME B46.1.
6. For design features purposes, this standard takes precedence over documents referenced herein.
7. Referenced documents shall be of the issue in effect on date of invitation for bid.

FIGURE 1. 90° elbow dimensions and configuration - Continued.

REQUIREMENTS:

Dimensions and configuration shall be in accordance with figure 1.

Materials and finishes shall be in accordance with SAE-AS4843, see table I for material code.

TABLE I. Material and finish code letters.

Material and finish code	Material	Protective chemical finish
B 1/	Copper alloy, type 377 forging in accordance with ASTM B124/B124M or half hard forging or bar in accordance with ASTM B138/B138M or bar in accordance with SAE-AMS4614.	No finish.
BC 1/ 3/	Copper alloy, type 377 forging in accordance with ASTM B124/B124M or half hard forging or bar in accordance with ASTM B138/B138M or bar in accordance with SAE-AMS4614.	Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 3.
D 2/	Aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 alloy 2014-T6. Aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6 alloy 2024-T6, or aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6 or SAE-AMS4339 alloy 2024-T851.	Anodize in accordance with SAE-AMS2472 or MIL-PRF-8625, type II, class 2, dye blue, duplex seal in accordance with procurement specification.
DV 2/	Aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 alloy 2014-T6. Aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6 alloy 2024-T6, or aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6 or SAE-AMS4339 alloy 2024-T851.	High purity aluminum in accordance with MIL-DTL-83488, class 3, type II with maximum coating thickness of .0005 inch. Glass bead peen pressure shall be 25 psi (1.72 bar) maximum.

See notes at end of table.

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TABLE I. Material and finish code - Continued.

Material and finish code	Material	Protective chemical finish
J	Type 304 corrosion resistant steel forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5639.	Passivate in accordance with SAE-AMS2700, method 1, type 6 or 7.
K	Type 316 corrosion resistant steel forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5648.	Passivate in accordance with SAE-AMS2700, method 1, type 6 or 7.
R	Type 321 corrosion resistant steel forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5645.	Passivate in accordance with SAE-AMS2700, method 1, type 6 or 7.
T 3/	Titanium in accordance with SAE-AMS4928 alloy 6Al-4V annealed or SAE-AMS4965	Anodize in accordance with SAE-AMS2488, type 2. 4/
TF 3/	Titanium in accordance with SAE-AMS4928 alloy 6Al-4V annealed.	Fluoride phosphate in accordance with SAE-AMS2486. 4/
W 2/	Aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 alloy 7075-T73 or aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/9 or SAE-AMS4124, alloy 7075-T7351 or 7075-T73.	Anodize in accordance with SAE-AMS2472 or MIL-PRF-8625, type II, class 2, dye brown similar to color 10080 in accordance with SAE-AMS-STD-595, duplex seal in accordance with procurement specification.
WV 2/	Aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 alloy 7075-T73 or aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/9 or SAE-AMS4124, alloy 7075-T7351 or 7075-T73.	High purity aluminum in accordance with MIL-DTL-83488, class 3, type II with maximum coating thickness of .0005 inch. Glass bead peen pressure shall be 25 psi (1.72 bar) maximum.

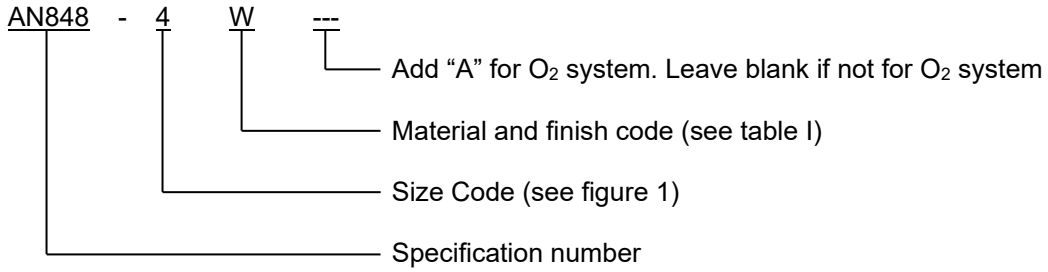
1/ Material code was dash on previous revisions, changed to agree with SAE-ARP1590.
2/ Inactive for new design, for new design use aluminum alloy 7075, code W, to improve stress corrosion resistance and tensile strength.
3/ Cadmium and titanium shall not be used in potable water or oxygen systems.
4/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to SAE-AMS-STD-595 colors 36076 through 36293.

Parts for use in oxygen systems (O_2) shall be identified in the Part or Identifying Number (PIN) as code “A” and shall be furnished cleaned, packaged, and labeled in accordance with SAE-AS611 to a process approved by the user.

Users are cautioned that the 90° elbow should be evaluated to their cleanliness requirements before installing in any equipment. Only qualified technical personnel with knowledge for the selection of cleaning methods for oxygen rich environments should make the determination as to what cleanliness level is acceptable for their application.

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PIN: The PIN consists of the prefix “AN” the specification sheet number, a dash number for hose and pipe thread size, material code and finish code letter, and oxygen code if applicable. Unassigned PIN’s shall not be used.



PIN examples:

AN848-8W indicates a 90° elbow hose to hose .500 inch (12.70 mm) hose to .375 inch (9.53 mm) pipe, aluminum alloy 7075-T73.

A848-8WV indicates a 90° elbow hose to hose .500 inch (12.70 mm) hose to .375 inch (9.53 mm) pipe, aluminum alloy 7075-T73 finish with high purity aluminum.

AN848-12WA indicates a 90° elbow hose to hose .750 inch (19.05 mm) hose to .750 inch (19.05 mm) pipe, aluminum alloy 7075-T73 for use in oxygen systems.

Guidance on use of alternative parts with less hazardous or non-hazardous materials. This specification provides for a number of alternative plating materials via the PIN. Users should select the PIN with the least hazardous material that meets the form, fit, and function requirements of their application.

Marking: Part shall be permanently marked with the AN PIN, and include the manufacturers CAGE, name, or trademark.

Inactive AN PIN and superseding AN PIN see table II.

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TABLE II. Superseding AN PIN.

Inactive AN PIN	Hose Size	Superseding AN PIN Inactive part
AN848-4	0.250	AN848-4B
AN848-4	0.250	AN848-4BC
AN848-4D	0.250	AN848-4W
AN848-4DV	0.250	AN848-4WV
AN848-6	0.375	AN848-6B
AN848-6	0.375	AN848-6BC
AN848-6D	0.375	AN848-6W
AN848-6DV	0.375	AN848-6WV
AN848-8	0.500	AN848-8B
AN848-8	0.500	AN848-8BC
AN848-8D	0.500	AN848-8W
AN848-8DV	0.500	AN848-8WV
AN848-10	0.625	AN848-10B
AN848-10	0.625	AN848-10BC
AN848-10D	0.625	AN848-10W
AN848-10DV	0.625	AN848-10WV
AN848-12	0.750	AN848-12B
AN848-12	0.750	AN848-12BC
AN848-12D	0.750	AN848-12W
AN848-12DV	0.750	AN848-12WV
AN848-16	1.000	AN848-16B
AN848-16	1.000	AN848-16BC
AN848-16D	1.000	AN848-16W
AN848-16DV	1.000	AN848-16WV
AN848-20	1.250	AN848-20B
AN848-20	1.250	AN848-20BC
AN848-20	1.250	AN848-20W
AN848-20	1.250	AN848-20WV
AN848-24	1.500	AN848-24B
AN848-24	1.500	AN848-24BC
AN848-24D	1.500	AN848-24W
AN848-24DV	1.500	AN848-24WV

Table III provides a detailed cross-reference of AN848 PINs and replacement SAE-AS5188 PINs. Users are cautioned to evaluate replacements for their particular application.

Cross reference data: SAE-AS5188 replacement numbers may be substituted for AN848 PINs, see table III.

CAUTION: The superseding information is valid as of the date of this specification and may be superseded by subsequent revisions of the superseding document.

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TABLE III. Cross-reference data. 1/

AN PIN Inactive part	Hose Size	AS PIN Active part	New design
AN848-4	0.250	AS5188B04 2/	
AN848-4BC	0.250	AS5188B04 2/	
AN848-4D	0.250	AS5188W04	AS5188W04
AN848-4DV	0.250	None	
AN848-4J	0.250	AS5188J04	
AN848-4K	0.250	AS5188K04	
AN848-4R	0.250	AS5188R04	
AN848-4S	0.250	AS5188R04	AS5188R04
AN848-4T	0.250	None	
AN848-4TF	0.250	None	
AN848-4W	0.250	AS5188W04	
AN848-4WV	0.250	None	
AN848-6	0.375	AS5188B06 2/	
AN848-6BC	0.375	AS5188B06 2/	
AN848-6D	0.375	AS5188W06	AS5188W06
AN848-6DV	0.375	None	
AN848-6J	0.375	AS5188J06	
AN848-6K	0.375	AS5188K06	
AN848-6R	0.375	AS5188R06	
AN848-6S	0.375	AS5188R06	AS5188R06
AN848-6T	0.375	None	
AN848-6TF	0.375	None	
AN848-6W	0.375	AS5188W06	
AN848-6WV	0.375	None	
AN848-8	0.500	AS5188B08 2/	
AN848-8BC	0.500	AS5188B08 2/	
AN848-8D	0.500	AS5188W08	AS5188W08
AN848-8DV	0.500	None	
AN848-8J	0.500	AS5188J08	
AN848-8K	0.500	AS5188K08	
AN848-8R	0.500	AS5188R08	
AN848-8S	0.500	AS5188R08	AS5188R08
AN848-8T	0.500	None	
AN848-8TF	0.500	None	
AN848-8W	0.500	AS5188W08	
AN848-8W	0.500	None	

See notes at end of table.

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TABLE III. Cross-reference data - Continued. 1/

AN PIN Inactive part	Hose Size	AS PIN Active part	New design
AN848-10	0.625	AS5188B10 2/	
AN848-10BC	0.625	AS5188B10 2/	
AN848-10D	0.625	AS5188W10	AS5188W10
AN848-10DV	0.625	None	
AN848-10J	0.625	AS5188J10	
AN848-10K	0.625	AS5188K10	
AN848-10R	0.625	AS5188R10	
AN848-10S	0.625	AS5188R10	AS5188R10
AN848-10T	0.625	None	
AN848-10TF	0.625	None	
AN848-10W	0.625	AS5188W10	
AN848-10WV	0.625	None	
AN848-12	0.750	AS5188B12 2/	
AN848-12BC	0.750	AS5188B12 2/	
AN848-12D	0.750	AS5188W12	AS5188W12
AN848-12DV	0.750	None	
AN848-12J	0.750	AS5188J12	
AN848-12K	0.750	AS5188K12	
AN848-12R	0.750	AS5188R12	
AN848-12S	0.750	AS5188R12	AS5188R12
AN848-12T	0.750	None	
AN848-12TF	0.750	None	
AN848-12W	0.750	AS5188W12	
AN848-12WV	0.750	None	
AN848-16	1.000	AN5188B16 2/	
AN848-16BC	1.000	AN5188B16 2/	
AN848-16D	1.000	AS5188W16	AS5188W16
AN848-16DV	1.000	None	
AN848-16J	1.000	AS5188J16	
AN848-16K	1.000	AS5188K16	
AN848-16R	1.000	AS5188R16	
AN848-16S	1.000	AS5188R16	AS5188R16
AN848-16T	1.000	None	
AN848-16TF	1.000	None	
AN848-16W	1.000	AS5188W16	
AN848-16WV	1.000	None	

See notes at end of table.

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TABLE III. Cross-reference data - Continued. 1/

AN PIN Inactive part	Hose Size	AS PIN Active part	New design
AN848-20	1.250	AS5188B20 2/	
AN848-20BC	1.250	AS5188B20 2/	
AN848-20D	1.250	AS5188W20	AS5188W20
AN848-20DV	1.250	None	
AN848-20J	1.250	AS5188J20	
AN848-20K	1.250	AS5188K20	
AN848-20R	1.250	AS5188R20	
AN848-20S	1.250	AS5188R20	AS5188R20
AN848-20T	1.250	None	
AN848-20TF	1.250	None	
AN848-20W	1.250	AS5188W20	
AN848-20WV	1.250	None	
AN848-24	1.500	AS5188B24 2/	
AN848-24BC	1.500	AS5188B24 2/	
AN848-24D	1.500	AS5188W24	AS5188W24
AN848-24D	1.500	None	
AN848-24J	1.500	AS5188J24	
AN848-24K	1.500	AS5188K24	
AN848-24R	1.500	AS5188R24	
AN848-24S	1.500	AS5188R24	AS5188R24
AN848-24T	1.500	None	
AN848-24TF	1.500	None	
AN848-24W	1.500	AS5188W24	
AN848-24WV	1.500	None	

1/ For new design use material designators R or W.

2/ SAE “B” designator for copper alloy parts can be either no finish or cadmium plate.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Referenced documents. In addition to SAE-AS4843, this document references the following:

MIL-PRF-8625	SAE-AMS-STD-595/36176	SAE-AMS2700
MIL-DTL-83488	SAE-AMS-STD-595/36231	SAE-AMS4124
ASME B46.1	SAE-AMS-STD-595/36251	SAE-AMS4339
ASTM B124/B124M	SAE-AMS-STD-595/36270	SAE-AMS4614
ASTM B138/B138M	SAE-AMS-STD-595/36280	SAE-AMS4928
SAE-AMS-STD-595/10080	SAE-AMS-STD-595/36293	SAE-AMS4965
SAE-AMS-STD-595/36076	SAE-AMS-QQ-A-225/6	SAE-AMS5639
SAE-AMS-STD-595/36081	SAE-AMS-QQ-A-225/9	SAE-AMS5645
SAE-AMS-STD-595/36099	SAE-AMS-QQ-A-367	SAE-AMS5648
SAE-AMS-STD-595/36118	SAE-AMS-QQ-P-416	SAE-ARP1590
SAE-AMS-STD-595/36134	SAE-AMS-QQ-S-763	SAE-AS611
SAE-AMS-STD-595/36152	SAE-AMS2472	SAE-AS5132
SAE-AMS-STD-595/36170	SAE-AMS2486	SAE-AS5188
SAE-AMS-STD-595/36173	SAE-AMS2488	SAE-AS8879

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2023-051)

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